

Course	Mandatory Activity	Skills Developed
Computer Networks	Interview-question quiz; Internal SIH participation	Networking fundamentals, troubleshooting, interview readiness, hackathon problem solving
Machine Learning Techniques	GATE practice; Research project literature study	ML concepts, analytical reasoning, research methodology, literature review, technical documentation
Advanced Data Structures & Algorithms	Practice problems, final test, Codethon, GATE practice	Advanced algorithms, complexity analysis, coding proficiency, competitive programming
Embedded System Design	IoT project with web development; National hackathon	Embedded programming, IoT, web integration, hardware-software integration, teamwork
OOP using Java	Practice problems, final test, Codethon, mini project	Java programming, OOP, application development, coding proficiency
Compiler Design	Quiz; Internal Hackathon	Compiler concepts, problem solving, technical implementation, teamwork
Operating Systems	GATE practice; industry-specific assessment	OS concepts, analytical thinking, technical aptitude, industry readiness
Database Management Systems	Practice/final test, GATE practice, mini project	SQL, database design, query development, DBMS problem solving, application development
Artificial Intelligence on Edge	Edge AI project; External hackathon	Edge AI, model deployment, embedded AI, real-time computing, innovation
Neural Networks & Deep Learning	NPTEL: Deep Learning for Computer Vision	Deep learning, computer vision, neural networks, research-oriented learning